FINANCIAL RISK ANALYSIS

Heavy Tails & Extreme Risk in Singapore's Banking Trio

A comparative study of DBS, OCBC and UOB daily stock returns

DBS	OCBC	UOB
D05.SI	O39.SI	U11.SI

DS603
Aryan Singh and Nikhil Balaji

Historical sample: 1 Jan 2014 – 31 Dec 2024 | DS603: Probability for Data Science

Project presentation

Why study extreme risk?

Financial losses are not always “normal”. Rare, unusually large market movements can matter disproportionately to investors and institutions.

01 How heavy are the tails?

Do extreme negative daily returns occur more often than a Normal model would suggest?

02 How large can losses be?

How do historical downside quantiles compare with fitted Normal and Student-t estimates?

03 Do banks fall together?

Are extreme-loss days shared across DBS, OCBC and UOB more often than under independence?

Project objectives

Compare: Normal vs Student-t return distributions.

Bound: Use Chebyshev’s and Cantelli’s inequalities for downside probabilities.

Examine: Correlation, joint crashes and conditional crash probabilities.

From stock prices to daily returns

Data source

Yahoo Finance via yfinance

DBS • D05.SI
OCBC • O39.SI
UOB • U11.SI

Period: 2014–2024
Frequency: daily trading observations

1
Download
Adjusted closing prices

→

2
Align
Keep common dates; handle missing rows

→

3
Transform
Calculate daily log returns

→

4
Analyse
Use the same return series across methods

Daily logarithmic return

$$r_t = \ln(P_t / P_{t-1})$$

Why returns? They put banks with different share prices on a comparable scale; log returns are additive over time.

Measuring the downside tail

Normal distribution

A familiar bell-shaped benchmark, described by its mean (μ) and standard deviation (σ).

Student-t distribution

Allows heavier tails when degrees of freedom are finite; useful when extreme returns are more common.

Maximum Likelihood

Estimate distribution parameters from observed returns, then calculate model-based probabilities and quantiles.

Value at Risk (VaR)

A downside quantile at a chosen probability.

Historical VaR

Observed return quantile

Normal VaR

Quantile from fitted Normal model

Student-t VaR

Quantile from fitted Student-t model

Interpretation: compare the models with the empirical tail. A fitted model is useful only if it represents the observed downside behaviour reasonably well.

Probability bounds and banks moving together

A. One-sided downside bounds

Chebyshev: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$

Cantelli: $P(X - \mu \leq -k\sigma) \leq 1/(1 + k^2)$

These are upper bounds, not estimates of the actual crash probability. Cantelli is tailored to a one-sided deviation.

B. Dependence and co-crashes

Pearson correlation: strength of linear co-movement.

Spearman correlation: monotonic association based on ranks.

Joint crash: $P(A \cap B)$
Conditional crash: $P(A \mid B) = P(A \cap B) / P(B)$

Compare observed joint crashes with $P(A) \times P(B)$, the independence benchmark.

Key caution: correlation does not equal crash probability. Historical co-crashes suggest dependence in the sample, not a guarantee about future events.

What this project can—and cannot—tell us

What the analysis provides

- Comparable historical returns for three banks
- Normal vs Student-t tail comparison
- Theoretical downside probability bounds
- Historical correlation and co-crash evidence

Limitations and extensions

- Historical data cannot guarantee future risk
- Extreme-tail estimates may be unstable
- No GARCH, EVT or copula model currently
- Extend with rolling windows, backtesting and tail-dependence methods

Next: live walkthrough of the Streamlit dashboard, Python modules and GitHub repository.

Thank you | Questions?